NSB BANGALORE

Master of Business Administration (MBA) - II Semester

Technology for Management (Course Code: 2.1)

PROJECT REPORT ON

HOTEL RESERVATION AND FRONT DESK MANAGEMENT SYSTEM

Design and Implementation using MySQL (RDBMS)

Submitted by

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CERTIFICATE

This is to certify that the project report titled "Hotel Reservation and Front Desk Management System" is a bonafide work carried out by the students named on the cover page, in partial fulfilment of the requirements of the Master of Business Administration (MBA) II Semester course Technology for Management (Course Code 2.1), Bangalore University, during the academic year 2026 - 2027.

The report has not been submitted earlier to any university or institution for the award of any degree, diploma or certificate.

Project Guide Head of the Department

DECLARATION

We hereby declare that this project report, including the MySQL scripts, diagrams and analysis, is our original work. Each member of the group has contributed to the modules mentioned in the Individual Attribution Log and is prepared to face an individual viva-voce on the contribution made. No part of this report has been copied from any other source without acknowledgement.

Place: Bengaluru Signature of the Students

Date:

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide and the faculty of the Department of Management Studies for their guidance during this project. We thank NSB Bangalore for including Technology for Management in the MBA curriculum, which gave us the opportunity to apply information-system concepts to a real business problem. We also thank our families and classmates for their support.

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1. INTRODUCTION AND PROBLEM STATEMENT

1.1 Background

Bengaluru is India's leading technology and start-up hub and receives a steady flow of business travellers, conference delegates, medical tourists and leisure guests throughout the year. Mid-sized business hotels in the city compete on speed of service, accuracy of billing and the ease with which a guest can book a room. In such an environment, the front desk is the nerve centre of the hotel: every reservation, arrival, room change, service charge and departure passes through it.

This project studies a hypothetical 12-room business hotel in Bengaluru and designs a computerised Hotel Reservation and Front Desk Management System (HRFMS). The system is built on a relational database (MySQL) and demonstrates how information systems convert day-to-day transactions into managerial information and decision support. The report follows the six modules of the syllabus.

1.2 Problem Statement

Hotels that still depend on paper registers and spreadsheets face recurring problems: double booking of rooms, delay in finding an available room, errors in manual bills, lost guest information, no record of which staff member handled which booking, and no reliable data for management to decide on pricing or marketing. The management needs one integrated, secure and accurate system.

1.3 Objectives

- To automate room reservation, check-in, service posting, billing, payment and check-out.
- To design a normalised (3NF) relational database that removes redundancy and protects data integrity.
- To provide MIS and DSS reports on occupancy, revenue, booking channels, guest loyalty and outstanding payments using SQL.
- To analyse the e-commerce, adoption, security, ethical and environmental aspects of the system.

1.4 Scope and Methodology

The scope covers the front-office function of a single hotel: guest registration, room inventory, reservations, stays, guest services and payments. Housekeeping, payroll and accounting are outside the scope and are treated only as external entities. The methodology consisted of (a) study of the hotel business process, (b) system analysis using a context diagram and Level-1 DFD, (c) database design using an ER diagram and normalisation, (d) implementation in MySQL with sample data, and (e) analysis through aggregate and sub-query reports. The sample data is fictitious and prepared only for this academic project; the date of reference for all outputs is 5 November 2026.

1.5 Syllabus Coverage

Syllabus Module	Project Deliverable	Chapter
1. IS Foundations	System classification (TPS, MIS, DSS, KMS, AI)	2
2. MIS and RDBMS	System view, ER design, 3NF schema	3
3. SAD and BPR	SDLC selection, BPR, Context diagram, Level-1 DFD	4
4. E-Commerce and Adoption	E-payment, security architecture, adoption model	5
5. IT Ethics and Security	Privacy, cybercrime, ergonomic and environmental audit	6
6. Application Software	Working MySQL scripts, DSS queries, executive report	7

Table 1: Mapping of syllabus modules to the project

2. MODULE 1: INFORMATION SYSTEM FOUNDATIONS AND CLASSIFICATION

2.1 Information System Components

An information system is an organised combination of hardware, software, data, people and procedures that collects, processes, stores and distributes information to support operations and decision making. In the HRFMS the hardware is the front-desk computers and a database server; the software is MySQL and the front-office application; the data are guest, room, reservation and payment records; the people are the front-desk staff, cashier and manager; and the procedures are the check-in, billing and check-out rules of the hotel.

2.2 Classification of the System

A modern hotel system is not a single type of information system. It is a combination of systems that serve different management levels. Table 2 classifies the HRFMS components.

System Type	Management Level	Function in HRFMS
TPS (Transaction Processing System)	Operational	Records reservations, check-in/check-out, service posting and payments; each transaction updates the database immediately.
MIS (Management Information System)	Middle management	Daily arrival list, room status summary, outstanding-payment report, staff-wise booking report.
DSS (Decision Support System)	Top management	Revenue by room type, channel-wise share, high-value guest analysis, comparison of bills with the average.
KMS (Knowledge Management System)	All levels	Guest preferences, standard operating procedures and complaint history stored for reuse.
AI application (future)	Strategic	Dynamic room pricing, demand forecasting, chatbot for booking enquiries.

Table 2: Classification of the system

Primary classification: The system is mainly a TPS at the front desk, which feeds an MIS and a DSS. The SQL queries in Chapter 7 act as the DSS layer because they combine data from many tables to answer management questions.

2.3 Enterprise Framework

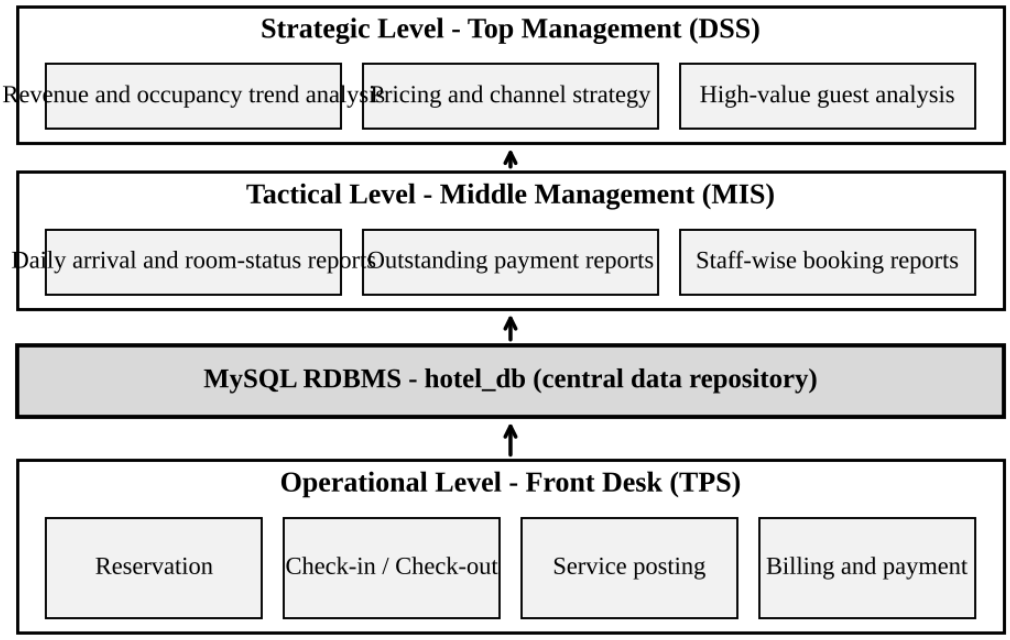


Figure 1: Enterprise framework of the HRFMS

3. MODULE 2: SYSTEM VIEW OF BUSINESS, MIS AND RDBMS DESIGN

3.1 System View of the Hotel Business

Element	Hotel Front Office
Inputs	Booking requests, guest identity proof, advance payments, service requests, room rates
Processing	Availability check, reservation, room allocation, check-in, service posting, billing, check-out
Outputs	Confirmation voucher, invoice, receipts, arrival lists, MIS and DSS reports
Feedback	Guest reviews, occupancy and revenue variance, complaints, used to revise rates and procedures
Environment	Guests, travel agents, payment gateways, regulators, competitors
Boundary	Front office; housekeeping, payroll and accounts are external

Table 3(a): System view of the business

Why an RDBMS. A relational database enforces integrity through primary keys, foreign keys and constraints, supports concurrent users and lets management ask new questions through SQL without changing the application.

3.2 Entity Relationship Diagram

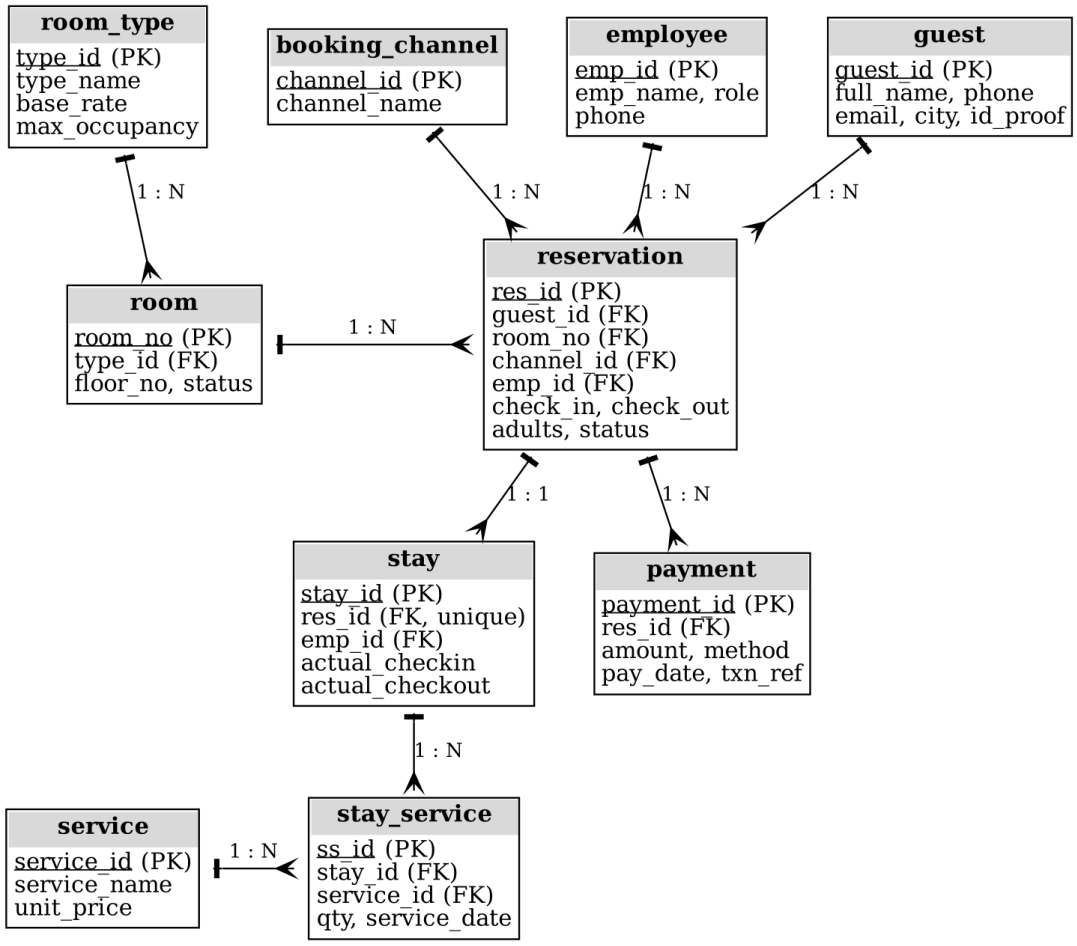


Figure 2: ER diagram of hotel_db (crow's-foot notation)

3.3 Entities and Relationships

Table	Purpose	Primary Key
guest	Customer identity and contact details	guest_id
room_type / room	Category with tariff; physical rooms and status	type_id / room_no
employee	Front-desk staff	emp_id
booking_channel	Walk-in, phone, website, OTA, corporate	channel_id
reservation	A guest's booking of a room for dates	res_id
stay	Actual check-in and check-out of a reservation	stay_id
service / stay_service	Chargeable services and their use during a stay	service_id / ss_id
payment	Advance and final payments	payment_id

Table 3: Entities of the database

3.4 Normalisation to Third Normal Form

Unnormalised design. A paper register records one row per booking with guest name, phone, city, room number, room type, tariff, channel name, employee name and payment details repeated in every row.

1NF. Every column holds a single value and each row is unique. Repeating groups such as several services or payments in one cell are moved to separate rows (stay_service and payment).

2NF. Partial dependencies on part of a key are removed. Guest details depend only on guest_id and not on the booking, so they are stored in guest. Service price depends only on service_id, so it is stored in service.

3NF. Transitive dependencies are removed. Tariff depends on room type, and room type depends on room number (room_no, type_id, base_rate). Storing base_rate in room would repeat it for every room of that type and risk inconsistency, so it is moved to room_type. Channel name and employee name are likewise moved to booking_channel and employee. After this step every non-key attribute depends on the key, the whole key and nothing but the key.

Integrity rules. Every table has a primary key; foreign keys enforce referential integrity (for example reservation.room_no must exist in room); ENUM lists and CHECK constraints control status, dates and amounts.

4. MODULE 3: SYSTEMS ANALYSIS AND DESIGN (SAD) AND BPR

4.1 SDLC Model Selection

Model	Strength	Limitation for this project
Waterfall	Simple, well documented	Rigid; requirements of a hotel change with guest behaviour
Prototyping	Early user feedback from front-desk staff	Needs disciplined documentation
Spiral	Strong risk analysis	Costly and complex for a small hotel
Agile / Iterative	Short cycles, continuous feedback	Needs active user involvement

Table 4: Comparison of SDLC models

Selected model: Iterative prototyping (agile-style) SDLC. The requirements are clear for reservation and billing but evolving for reporting. The database is built first as a working prototype, front-desk staff review the screens and reports, and the next iteration adds features such as online payments. Phases followed: (1) planning and feasibility, (2) analysis, (3) design, (4) implementation, (5) testing, and (6) maintenance.

Feasibility. The project is technically feasible (open-source MySQL on ordinary hardware), economically feasible (no licence cost, savings from fewer billing errors) and operationally feasible with staff training (Chapter 5).

4.2 Business Process Re-engineering (BPR)

Process	As-Is (Manual)	To-Be (HRFMS)
Availability check	Staff scan the register; 5 to 10 minutes	Single SQL query; a few seconds
Reservation	Paper entry, double booking possible	Validated entry with foreign keys and date check
Check-in	Re-writing guest details	Details fetched from guest table; status updated in one transaction
Billing	Manual totals, tax errors	Invoice view calculates room, service and tax automatically
Reporting	Month-end compilation by hand	On-demand MIS and DSS queries

Table 5: BPR - as-is and to-be processes

4.3 Context Diagram (Level 0)

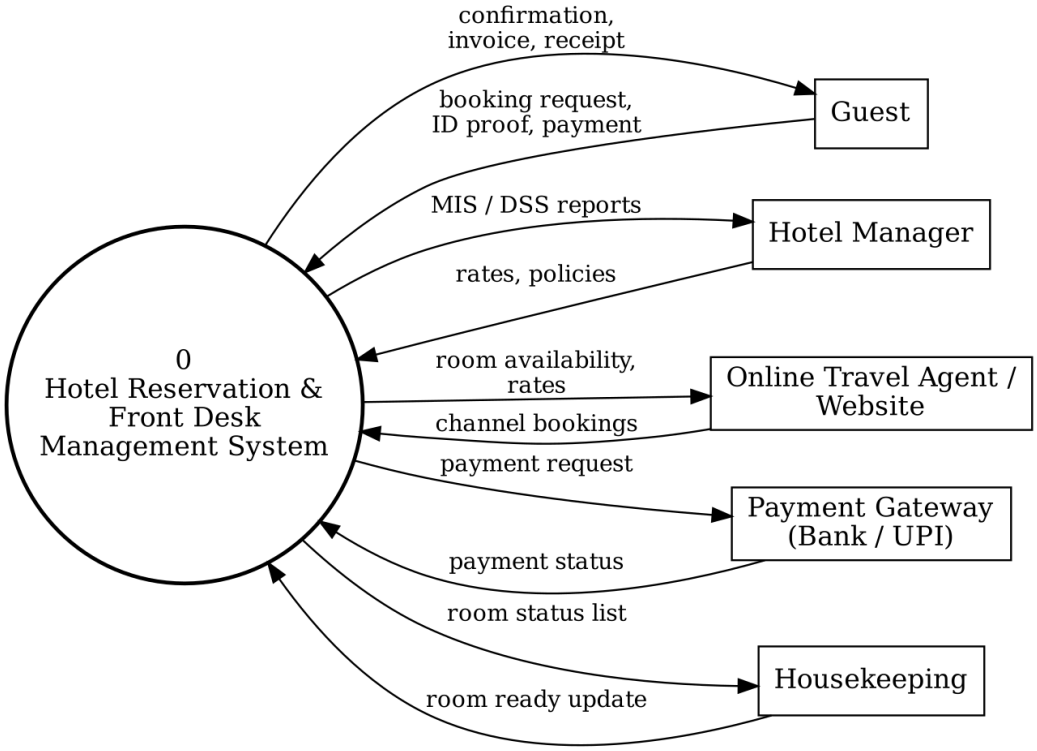


Figure 3: Context diagram of the HRFMS

The context diagram shows the whole system as one process and its data exchange with five external entities: the Guest, the Hotel Manager, the Online Travel Agent/Website, the Payment Gateway and Housekeeping.

4.4 Level-1 Data Flow Diagram

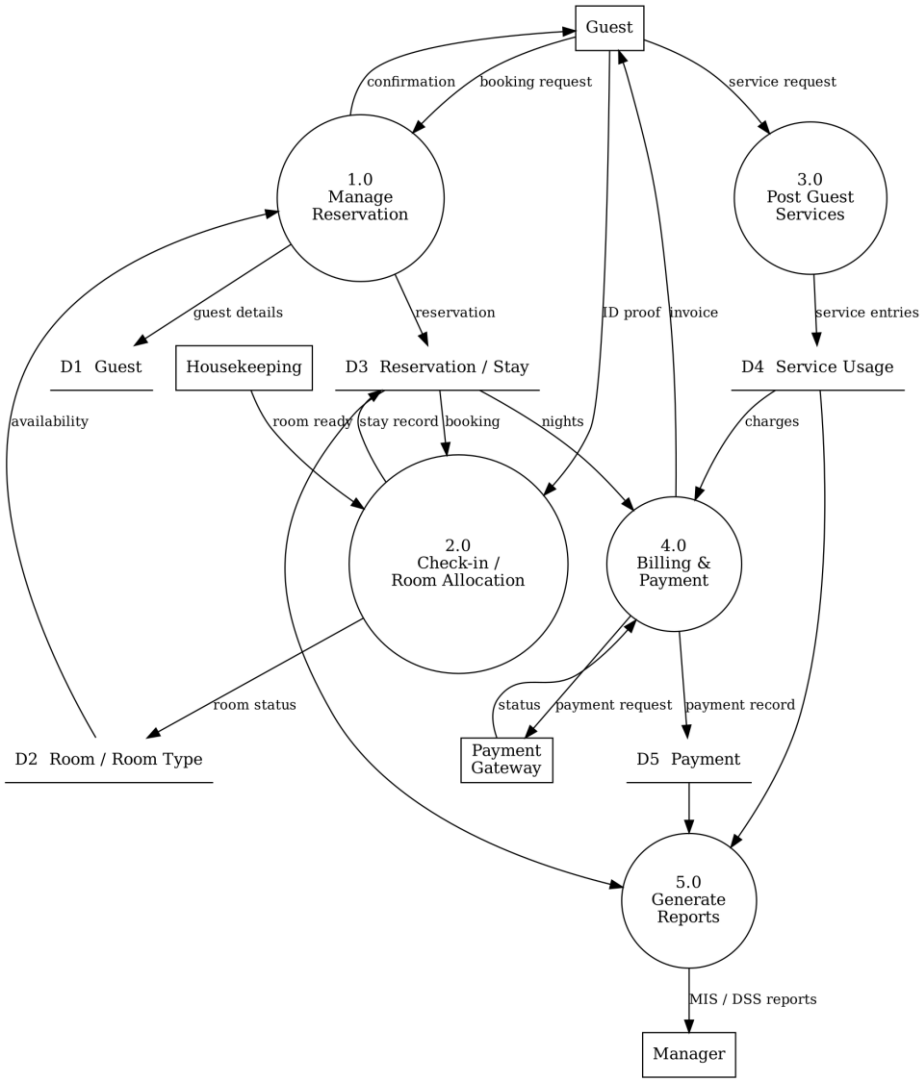


Figure 4: Level-1 DFD (D1 to D5 are data stores)

The Level-1 DFD decomposes the system into five processes: Manage Reservation, Check-in and Room Allocation, Post Guest Services, Billing and Payment, and Generate Reports. Each process reads and writes the data stores that correspond to tables of the database.

5. MODULE 4: E-COMMERCE, E-PAYMENT AND TECHNOLOGY ADOPTION

5.1 E-Commerce Model

The HRFMS supports a B2C e-commerce channel through the hotel website and a B2B channel through corporate and online-travel-agent bookings. Online booking increases reach, reduces commission paid to intermediaries and gives the hotel direct ownership of guest data. In the sample data, the hotel website is the largest single channel with 35.7 per cent of non-cancelled bookings (see Query 2).

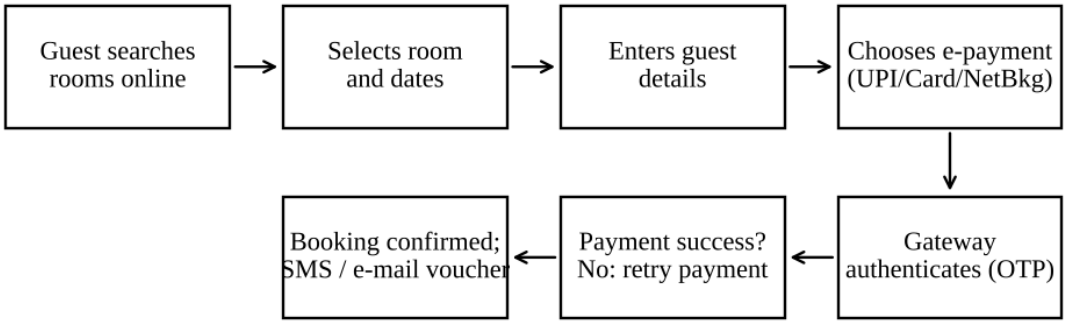


Figure 5: Process map of the online booking and payment

5.2 E-Payment Integration

Method	How it works	Control
UPI	Guest pays through a UPI app using a virtual payment address or QR code	Bank PIN, transaction reference stored in txn_ref
Debit / Credit card	Card details entered on the gateway page	3-D Secure OTP, card data tokenised
Net banking	Redirect to the bank's own secure page	Bank authentication
Cash at desk	Counter payment with printed receipt	Receipt number, cashier accountability

Table 6: Payment methods supported

The hotel does not store card numbers. The payment gateway returns only a status and a reference number, which are recorded in the payment table. This reduces the hotel's exposure under PCI-DSS requirements.

5.3 Security Architecture

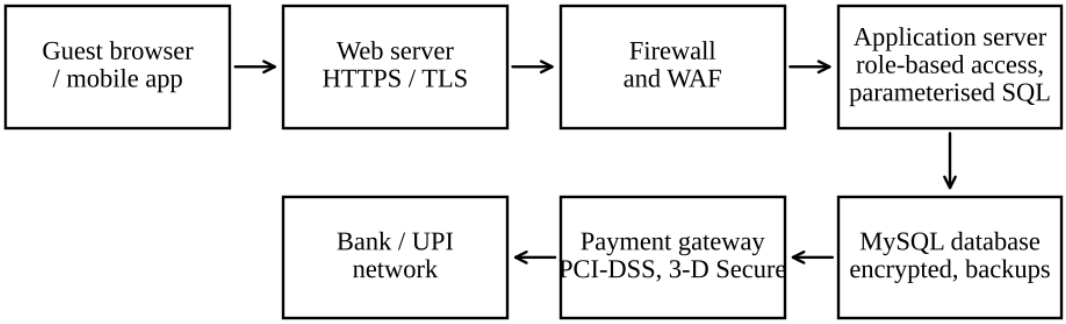


Figure 6: Security architecture for online booking

Security is layered. Traffic is encrypted by HTTPS, a firewall filters requests, the application enforces role-based access, database queries are parameterised to prevent SQL injection, and sensitive data are encrypted and backed up regularly.

5.4 Technology Adoption

The success of the system depends on its acceptance by users. Two frameworks are applied. The Technology Acceptance Model (TAM) states that adoption depends on perceived usefulness and perceived ease of use. Rogers' Diffusion of Innovations classifies adopters into innovators, early adopters, early majority, late majority and laggards.

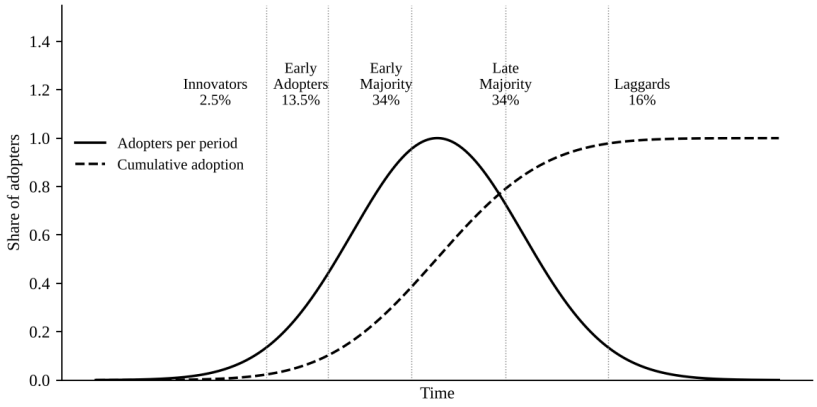


Figure 7: Technology adoption curve (Rogers)

- Innovators and early adopters (front desk manager, reservation executive): pilot the system for two weeks and give feedback.
- Early majority (receptionists, cashier): hands-on training sessions and a one-page quick guide.
- Late majority and laggards (senior staff): peer support, parallel run of the paper register for one month, and visible management support.
- Guests: simple online booking form, UPI payment and instant e-mail voucher to encourage use.

6. MODULE 5: IT ETHICS, SECURITY AND AUDIT

6.1 Data Privacy

The hotel collects personal data such as names, phone numbers, e-mail addresses and identity proof numbers. This data must be collected only for a stated purpose, kept accurate, protected from unauthorised access and deleted when no longer needed. The relevant Indian laws are the Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023 together with its rules. Hotels must also maintain guest records as required by local police and, for foreign nationals, the prescribed arrival-reporting rules. The following controls are applied in the design:

- Consent notice on the booking form and the registration card.
- Role-based access: receptionists see guest and booking data; only the manager sees financial reports.
- Identity proof numbers stored in a restricted column and masked on printed documents.
- Encrypted backups, audit logging of changes and a retention schedule.
- Data of cancelled or old bookings anonymised after the legal retention period.

6.2 Cybercrime Risk Matrix

Risk	Likelihood	Impact	Mitigation
SQL injection on booking form	Medium	High	Parameterised queries, input validation
Phishing of staff credentials	High	High	Awareness training, two-factor login
Payment fraud / card theft	Medium	High	Gateway tokenisation, 3-D Secure
Ransomware	Low	High	Offline backups, patching, antivirus
Insider misuse of guest data	Medium	Medium	Role-based access, audit logs
Fake OTA / website	Medium	Medium	Official domain, SSL certificate, guest advisory

Table 7: Cybercrime risk matrix

Under the IT Act, 2000, offences such as unauthorised access, data theft, identity theft and cheating by personation using a computer resource are punishable, and companies that fail to protect sensitive personal data may be liable to pay compensation. Staff must be informed of these consequences.

6.3 Ethical, Ergonomic and Environmental Audit

Area	Observation / Requirement	Action
Ethical use	Guest data used only for hotel service	Written data-use policy; staff sign confidentiality undertaking

Area	Observation / Requirement	Action
Accuracy	Wrong bills harm guests	Automated invoice view; manager approval for waivers
Ergonomics	Reception staff sit and type for long hours	Adjustable chair, monitor at eye level, 20-20-20 eye rest rule
Lighting and glare	Screen glare at the counter	Anti-glare screens, indirect lighting
Environment	Paper registers and printed bills	Paperless registration, e-invoice by mail
E-waste	Old PCs and printers	Authorised e-waste recyclers
Energy	Always-on equipment	Energy-saving power plans; shut down after shift

Table 8: Ethics, ergonomic and environmental audit

7. MODULE 6: APPLICATION SOFTWARE - MYSQL IMPLEMENTATION

This chapter presents the working database. All code was executed on a MySQL-compatible server and the outputs shown are the actual results. The complete script is submitted as the file hotel_management_system.sql. Sample data is fictitious.

7.1 Creating the Database and Tables (DDL)

Input (SQL):

```
CREATE TABLE guest (
  guest_id    INT AUTO_INCREMENT PRIMARY KEY,
  full_name   VARCHAR(60) NOT NULL,
  phone       VARCHAR(15) NOT NULL UNIQUE,
  email       VARCHAR(80),
  city        VARCHAR(40),
  id_proof    VARCHAR(30) NOT NULL
);
CREATE TABLE room_type (
  type_id       INT AUTO_INCREMENT PRIMARY KEY,
  type_name     VARCHAR(30) NOT NULL UNIQUE,
  base_rate     DECIMAL(8,2) NOT NULL CHECK (base_rate > 0),
  max_occupancy INT NOT NULL
);
CREATE TABLE room (
  room_no   INT PRIMARY KEY,
  type_id   INT NOT NULL,
  floor_no  INT NOT NULL,
  status    ENUM('Available','Occupied','Cleaning','Maintenance') DEFAULT 'Available',
  FOREIGN KEY (type_id) REFERENCES room_type(type_id)
);
CREATE TABLE employee (
  emp_id    INT AUTO_INCREMENT PRIMARY KEY,
  emp_name  VARCHAR(60) NOT NULL,
  role      VARCHAR(40) NOT NULL,
  phone     VARCHAR(15)
);
CREATE TABLE booking_channel (
  channel_id   INT AUTO_INCREMENT PRIMARY KEY,
  channel_name VARCHAR(30) NOT NULL UNIQUE
);
```

Input (SQL):

```
CREATE TABLE reservation (
  res_id     INT AUTO_INCREMENT PRIMARY KEY,
  guest_id   INT NOT NULL,
  room_no    INT NOT NULL,
  channel_id INT NOT NULL,
  emp_id     INT NOT NULL,
  check_in   DATE NOT NULL,
```

```
  check_out  DATE NOT NULL,
  adults     INT NOT NULL DEFAULT 1,
  status     ENUM('Confirmed','Checked-in','Checked-out','Cancelled') DEFAULT
  'Confirmed',
  CHECK (check_out > check_in),
  FOREIGN KEY (guest_id)   REFERENCES guest(guest_id),
  FOREIGN KEY (room_no)    REFERENCES room(room_no),
  FOREIGN KEY (channel_id) REFERENCES booking_channel(channel_id),
  FOREIGN KEY (emp_id)     REFERENCES employee(emp_id)
);
CREATE TABLE stay (
  stay_id        INT AUTO_INCREMENT PRIMARY KEY,
  res_id         INT NOT NULL UNIQUE,
  actual_checkin  DATETIME NOT NULL,
  actual_checkout DATETIME NULL,
  emp_id         INT NOT NULL,
  FOREIGN KEY (res_id) REFERENCES reservation(res_id),
  FOREIGN KEY (emp_id) REFERENCES employee(emp_id)
);
CREATE TABLE service (
  service_id   INT AUTO_INCREMENT PRIMARY KEY,
  service_name VARCHAR(40) NOT NULL UNIQUE,
  unit_price   DECIMAL(8,2) NOT NULL
);
CREATE TABLE stay_service (
  ss_id       INT AUTO_INCREMENT PRIMARY KEY,
  stay_id     INT NOT NULL,
  service_id  INT NOT NULL,
  qty         INT NOT NULL DEFAULT 1,
  service_date DATE NOT NULL,
  FOREIGN KEY (stay_id)    REFERENCES stay(stay_id),
  FOREIGN KEY (service_id) REFERENCES service(service_id)
);
CREATE TABLE payment (
  payment_id INT AUTO_INCREMENT PRIMARY KEY,
  res_id     INT NOT NULL,
  amount     DECIMAL(10,2) NOT NULL CHECK (amount > 0),
  method     ENUM('Cash','Card','UPI','NetBanking') NOT NULL,
  pay_date   DATE NOT NULL,
  txn_ref    VARCHAR(30),
  FOREIGN KEY (res_id) REFERENCES reservation(res_id)
);
```

7.2 Inserting Data (Input)

Input (SQL):

```
INSERT INTO room_type (type_name, base_rate, max_occupancy) VALUES
 ('Standard', 2500, 2), ('Deluxe', 3800, 3),
 ('Executive Suite', 6500, 4), ('Premium Suite', 9000, 4);
INSERT INTO room (room_no, type_id, floor_no, status) VALUES
```

Room, employee, channel and service data were inserted in the same way. Guest and reservation data are shown below in abbreviated form; the full list is in the submitted .sql file.

Input (SQL):

```
INSERT INTO guest (full_name, phone, email, city, id_proof) VALUES
 ('Ramesh Kumar','9886100001','ramesh.k@mail.com','Bengaluru','Aadhaar'),
 ('Anita Sharma','9886100002','anita.s@mail.com','New Delhi','Passport'), ...;
INSERT INTO reservation (guest_id, room_no, channel_id, emp_id, check_in, check_out,
adults, status) VALUES
 (1,101,3,2,'2026-10-20','2026-10-23',2,'Checked-out'),
 (2,201,4,4,'2026-10-22','2026-10-25',2,'Checked-out'), ...;
```

7.3 Invoice View

A view named v_invoice calculates, for every reservation, the room charges, service charges, GST at a flat 12 per cent (assumed for simplicity; actual rates depend on the tariff slab and the current GST notification), total bill and amount paid. It is used by the billing and DSS queries.

Input (SQL):

```
CREATE VIEW v_invoice AS
SELECT r.res_id, g.full_name, r.room_no, rt.type_name,
       DATEDIFF(r.check_out, r.check_in)                    AS nights,
       DATEDIFF(r.check_out, r.check_in) * rt.base_rate      AS room_charges,
       COALESCE(sv.svc, 0)                                  AS service_charges,
       ROUND(0.12 * (DATEDIFF(r.check_out, r.check_in) * rt.base_rate
                  + COALESCE(sv.svc, 0)), 2)                AS gst_12,
       ROUND(1.12 * (DATEDIFF(r.check_out, r.check_in) * rt.base_rate
                  + COALESCE(sv.svc, 0)), 2)                AS total_bill,
       COALESCE(pm.paid, 0)                                 AS paid
FROM reservation r
JOIN guest g       ON g.guest_id = r.guest_id
JOIN room rm       ON rm.room_no = r.room_no
JOIN room_type rt  ON rt.type_id = rm.type_id
LEFT JOIN (SELECT st.res_id, SUM(ss.qty * s.unit_price) AS svc
           FROM stay st
           JOIN stay_service ss ON ss.stay_id = st.stay_id
           JOIN service s       ON s.service_id = ss.service_id
           GROUP BY st.res_id) sv ON sv.res_id = r.res_id
LEFT JOIN (SELECT res_id, SUM(amount) AS paid
           FROM payment GROUP BY res_id) pm ON pm.res_id = r.res_id
WHERE r.status <> 'Cancelled';
```

Input (SQL):

```
INSERT INTO payment (res_id, amount, method, pay_date, txn_ref) VALUES
 (1,2000,'UPI','2026-10-10','UPI88120011'), ...;
INSERT INTO payment (res_id, amount, method, pay_date, txn_ref)
SELECT i.res_id, i.total_bill - i.paid, 'Card', DATE(s.actual_checkout),
CONCAT('SETTLE-', i.res_id)
```

```
FROM v_invoice i JOIN stay s ON s.res_id = i.res_id
WHERE s.actual_checkout IS NOT NULL AND i.total_bill > i.paid;
```

7.4 Front Desk Operations (Input and Output)

Operation 1 - Create a new reservation with an advance payment.

Input (SQL):

```
INSERT INTO reservation (guest_id, room_no, channel_id, emp_id, check_in, check_out,
adults)
VALUES (4, 202, 2, 4, '2026-11-20', '2026-11-22', 1);
INSERT INTO payment (res_id, amount, method, pay_date, txn_ref)
VALUES (LAST_INSERT_ID(), 2000, 'UPI', '2026-11-05', 'UPI88120150');
```

Output:

res_id	full_name	room_no	check_in	check_out	status
15	Priya Nair	202	2026-11-20	2026-11-22	Confirmed

Operation 2 - Check-in of reservation 10 (a transaction updates three tables together).

Input (SQL):

```
START TRANSACTION;
INSERT INTO stay (res_id, actual_checkin, emp_id) VALUES (10, '2026-11-05 13:30:00',
2);
UPDATE reservation SET status = 'Checked-in' WHERE res_id = 10;
UPDATE room SET status = 'Occupied' WHERE room_no = 201;
COMMIT;
```

Output:

res_id	room_no	res_status	room_status	actual_checkin
10	201	Checked-in	Occupied	2026-11-05 13:30:00

Operation 3 - Post services to a guest's stay and view the bill.

Input (SQL):

```
INSERT INTO stay_service (stay_id, service_id, qty, service_date)
VALUES (9, 6, 2, '2026-11-05');
```

Output:

stay_id	service_name	qty	unit_price	amount
9	Minibar	3	300.00	900.00
9	Room Service	2	600.00	1200.00

Input (SQL):

```
SELECT full_name, room_charges, service_charges, total_bill,
       paid, total_bill - paid AS balance
FROM v_invoice WHERE res_id = 9;
```

Output:

full_name	room_charges	service_charges	total_bill	paid	balance
Deepa Iyer	5000.00	2100.00	7952.00	2000.00	5952.00

Operation 4 - Check-out with final settlement.

Input (SQL):

```
START TRANSACTION;
INSERT INTO payment (res_id, amount, method, pay_date, txn_ref)
SELECT res_id, total_bill - paid, 'UPI', '2026-11-05', 'UPI88120160'
FROM v_invoice WHERE res_id = 9;
UPDATE stay SET actual_checkout = '2026-11-05 11:20:00' WHERE res_id = 9;
UPDATE reservation SET status = 'Checked-out' WHERE res_id = 9;
UPDATE room SET status = 'Cleaning' WHERE room_no = 103;
COMMIT;
```

Output:

res_id	res_status	room_status	actual_checkout	balance_due
9	Checked-out	Cleaning	2026-11-05 11:20:00	0.00

After checkout the balance due is zero and the room status changes to Cleaning, so the housekeeping team knows that the room is to be prepared for the next guest.

7.5 Decision Support (DSS) Queries

Query 1 - Revenue by room type (JOIN, GROUP BY).

Input (SQL):

```
SELECT rt.type_name, COUNT(*) AS bookings,
       SUM(DATEDIFF(r.check_out, r.check_in)) AS room_nights,
       SUM(DATEDIFF(r.check_out, r.check_in) * rt.base_rate) AS room_revenue
FROM reservation r
JOIN room rm      ON rm.room_no = r.room_no
JOIN room_type rt ON rt.type_id = rm.type_id
WHERE r.status IN ('Checked-in','Checked-out')
GROUP BY rt.type_name
ORDER BY room_revenue DESC;
```

Output:

type_name	bookings	room_nights	room_revenue
Deluxe	4	13	49400.00

type_name	bookings	room_nights	room_revenue
Executive Suite	2	7	45500.00
Premium Suite	1	3	27000.00
Standard	3	7	17500.00

Query 2 - Booking channel analysis (sub-query for percentage).

Input (SQL):

```
SELECT bc.channel_name, COUNT(*) AS bookings,
       ROUND(100 * COUNT(*) / (SELECT COUNT(*) FROM reservation
                               WHERE status <> 'Cancelled'), 1) AS pct_share
FROM reservation r
JOIN booking_channel bc ON bc.channel_id = r.channel_id
WHERE r.status <> 'Cancelled'
GROUP BY bc.channel_name
ORDER BY bookings DESC;
```

Output:

channel_name	bookings	pct_share
Website	5	35.7
Online Travel Agent	3	21.4
Phone	3	21.4
Corporate	2	14.3
Walk-in	1	7.1

Query 3 - Repeat guests (GROUP BY with HAVING).

Input (SQL):

```
SELECT g.full_name, g.city, COUNT(*) AS stays
FROM reservation r JOIN guest g ON g.guest_id = r.guest_id
WHERE r.status <> 'Cancelled'
GROUP BY g.guest_id, g.full_name, g.city
HAVING COUNT(*) > 1;
```

Output:

full_name	city	stays
Ramesh Kumar	Bengaluru	2
Anita Sharma	New Delhi	2
Priya Nair	Kochi	2
Mohammed Imran	Hyderabad	2

Query 4 - Service-wise revenue.

Input (SQL):

```
SELECT s.service_name, SUM(ss.qty) AS units_sold,
       SUM(ss.qty * s.unit_price) AS revenue
FROM stay_service ss JOIN service s ON s.service_id = ss.service_id
GROUP BY s.service_name
ORDER BY revenue DESC;
```

Output:

service_name	units_sold	revenue
Breakfast Buffet	18	8100.00
Spa Session	4	8000.00
Room Service	11	6600.00
Airport Pickup	2	2400.00
Minibar	5	1500.00
Laundry	7	1050.00

Query 5 - Guests whose bill is above the average bill (sub-query).

Input (SQL):

```
SELECT full_name, room_no, total_bill
FROM v_invoice
WHERE total_bill > (SELECT AVG(total_bill) FROM v_invoice)
ORDER BY total_bill DESC;
```

Output:

full_name	room_no	total_bill
Kavitha Reddy	401	37408.00
John Mathew	302	31808.00
Suresh Babu	301	28336.00
Lakshmi Narayan	402	20160.00
Ramesh Kumar	203	19376.00

Query 6 - Outstanding payments.

Input (SQL):

```
SELECT res_id, full_name, total_bill, paid, total_bill - paid AS outstanding
FROM v_invoice
WHERE total_bill > paid
ORDER BY outstanding DESC;
```

Output:

res_id	full_name	total_bill	paid	outstanding
8	John Mathew	31808.00	8000.00	23808.00

res_id	full_name	total_bill	paid	outstanding
11	Lakshmi Narayan	20160.00	2500.00	17660.00
7	Ramesh Kumar	19376.00	4000.00	15376.00
10	Arjun Gowda	12768.00	3000.00	9768.00
13	Mohammed Imran	8512.00	0.00	8512.00
15	Priya Nair	8512.00	2000.00	6512.00
12	Anita Sharma	5600.00	2000.00	3600.00

Query 7 - Executive summary of billing.

Input (SQL):

```
SELECT ROUND(SUM(room_charges), 2)   AS room_revenue,
       ROUND(SUM(service_charges), 2) AS service_revenue,
       ROUND(SUM(gst_12), 2)          AS gst_collected,
       ROUND(SUM(total_bill), 2)      AS gross_billing
FROM v_invoice;
```

Output:

room_revenue	service_revenue	gst_collected	gross_billing
177600.00	27650.00	24630.00	229880.00

7.6 Executive Report - Interpretation

- **Revenue mix:** Deluxe rooms earned the highest room revenue (Rs. 49,400) because of more room-nights, while the Premium Suite earned Rs. 27,000 from only one booking. Suites give high revenue per night and deserve targeted promotion to corporate clients.
- **Channels:** The hotel website brings 35.7 per cent of bookings, more than any other channel. Online travel agents bring 21.4 per cent and charge commission, so shifting guests to the direct website would improve margin.
- **Services:** Breakfast and spa are the largest ancillary earners (Rs. 8,100 and Rs. 8,000). Packages that bundle these with rooms can raise the average bill.
- **Loyalty:** Four guests have stayed more than once; they are candidates for a loyalty programme.
- **Cash flow:** Outstanding payments (Query 6) should be followed up before the stay ends, and the advance should be raised for high-value bookings.

Note: gross billing in Query 7 includes confirmed future bookings and is therefore a billing forecast, not realised income.

8. FINDINGS, CONCLUSION AND FUTURE SCOPE

8.1 Findings

- A normalised database removed repetition of guest, tariff, channel and employee details and prevented inconsistent data.
- Transactions for check-in and check-out kept reservation, stay, room and payment tables consistent.
- SQL aggregate queries gave management immediate answers that earlier needed hours of manual work.
- Direct online bookings and ancillary services are the main levers for improving profitability.

8.2 Conclusion

The Hotel Reservation and Front Desk Management System demonstrates the application of all six modules of Technology for Management: classification of the system as TPS, MIS and DSS; a system view and 3NF relational design; SDLC, BPR and data-flow modelling; e-commerce, payment security and adoption planning; privacy, cybercrime and ergonomic considerations; and a working MySQL application. The project shows that a well-designed database is the foundation on which operational efficiency and managerial decision making are built.

8.3 Limitations and Future Scope

- The database uses a current tariff and does not store historical rate changes or seasonal rates.
- Future work includes a web or mobile front end, integration with a live payment gateway, housekeeping and accounting modules, and AI-based demand forecasting for dynamic pricing.

9. REFERENCES AND INDIVIDUAL ATTRIBUTION LOG

9.1 Individual Attribution Log

As per the project guidelines, each student owns a module matrix. Replace the placeholders with the names of the group members.

Student	Modules Owned	Contribution
Student 1 (Data and RDBMS Engineer)	2 and 6	ER diagram, 3NF schema, MySQL scripts, DSS queries
Student 2 (Systems Analyst)	1 and 3	System classification, SDLC, BPR, context diagram, DFD
Student 3 (E-Commerce and Ethics Specialist)	4 and 5	Payment and security framework, adoption analysis, ethics audit, report consolidation

Table 9: Individual attribution log (3-member group)

9.2 References

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